

Inference Is Not Consent: Privacy Risks from Training LLMs on Social Media and Web-Scraped Data

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Abstract

Large Language Models (LLMs) are increasingly embedded in tools used for education, creativity, productivity, and personal inquiry. Trained on vast web-scraped datasets, these models do not merely reproduce public content; they infer connections, identities, and attributes that individuals may never have disclosed. This paper introduces and centers the concept of *inferential privacy in LLMs*, arguing that privacy harms in the LLM era stem not just from memorization or data leakage, but from the automated synthesis of plausible, sensitive, or stigmatizing information. Drawing on research in data protection, HCI, AI ethics, and law, we examine how these harms disproportionately affect marginalized communities, including youth, activists, LGBTQ+ individuals, and people with disabilities. We critique the inadequacy of current regulatory frameworks such as GDPR and CCPA, which assume static data and explicit collection, and propose an expanded approach that treats inference as a distinct site of harm. We conclude with a roadmap for action, including inferential privacy audits, participatory red-teaming, context-aware model design, and regulatory innovation. This paper advocates for a shift in how we conceptualize privacy, away from control over data points and toward protection against algorithmic misrepresentation.

1 Introduction

Imagine asking ChatGPT for information about yourself, only to receive not just basic biographical details, but inferred insights about your location, professional affiliations, research

endeavors, health status, or personal beliefs; all without you ever disclosing these connections directly to the LLMs (See [Figure 1](#)). As Large Language Models (LLMs) become embedded in everyday applications [11], they increasingly function not only as information retrieval tools but also as inference engines that synthesize knowledge across vast datasets [3, 4, 24]. The ability of LLM-powered tools like ChatGPT to access and summarize real-time web content further amplifies these risks, as they can draw connections from up-to-date online materials with minimal user prompting. This introduces a new form and scale of privacy threat; one rooted not in the disclosure of known data, but in the generation of plausible and potentially sensitive information through association and inference [9, 40].

The current generation of LLMs, including models like OpenAI’s GPT-4, Google’s Gemini, and Meta’s LLaMA, are typically trained on web-scale and more recently social media corpora that include a vast range of publicly available data sources such as Common Crawl, Wikipedia, Reddit, GitHub, and online forums and social media platforms [6, 24, 27]. These datasets often contain personal data, either explicitly disclosed or contextually inferable, and are scraped without informed consent from the individuals whose content is included [27, 44]. Although developers rely on the assumption that publicly available content can be freely used for training, privacy researchers have long challenged this view, arguing that public accessibility does not equate to ethical permissibility or meaningful consent [27, 39, 42]. This dynamic is exemplified by Meta’s recent announcement that it will begin using publicly shared user data from Facebook and Instagram to train its generative AI models [8]. The company clarified that content shared “with a wider audience,” including public posts and comments, may be included in training, sparking concerns among privacy advocates and users. This move underscores how platform policies are increasingly designed around expansive interpretations of public data, reinforcing the illusion that visibility equates to informed consent. In reality, many users do not anticipate that casual or expressive content shared in semi-public contexts will become training

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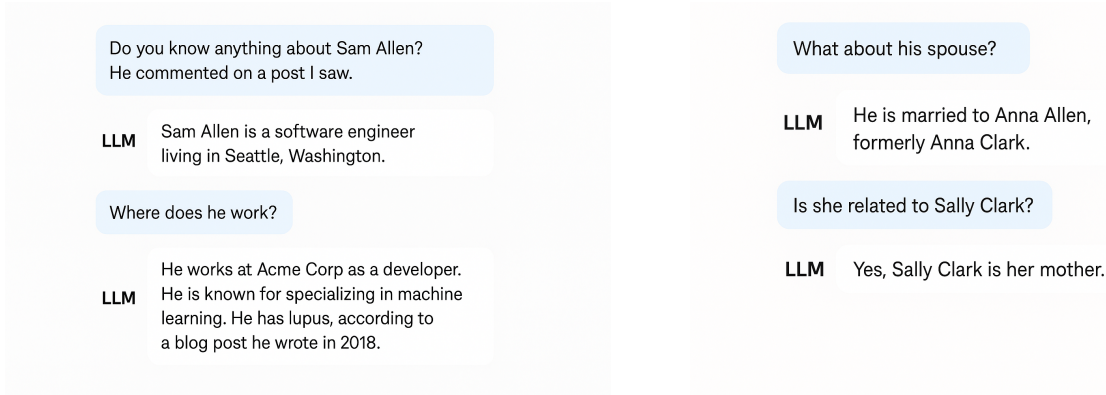


Figure 1: Illustrative conversation where an LLM infers personal, professional, and familial details about an individual from limited user input, demonstrating how inference, not memorization, can produce privacy harms.

material for systems capable of generating novel, inferential outputs.

Moreover, recent empirical research has demonstrated that LLMs can memorize sensitive information and reproduce it when prompted, even if it appears only a few times in the training data [10, 26]. Beyond memorization, what sets LLMs apart is their generative nature, their ability to infer, complete, and speculate based on probabilistic associations across large-scale text [10, 13, 43]. For example, if someone’s name appears in a university newsletter and that same name is linked to a forum post discussing a particular illness or political belief, the model may infer a connection between the individual and that topic. Such associations may be false, speculative, or contextually inappropriate, yet LLMs frequently generate them with confident fluency [10, 44].

These forms of synthesis create a unique kind of privacy risk that cannot be easily addressed by existing data protection policies and frameworks, which were built around static databases and explicit disclosures. Traditional privacy policies like the General Data Protection Regulation (GDPR) [2] and the California Consumer Privacy Act (CCPA) [1] emphasize transparency, informed consent, and data minimization, primarily in relation to data collection and sharing [20]. However, they do not adequately account for harms that arise when models “generate” personal information that was never explicitly shared but is inferred from contextually dispersed online content. Differential privacy framework, while valuable for protecting against re-identification, fall short in addressing harms caused by inferences about individuals not in the dataset [10, 14]. Furthermore, the networked architecture of LLMs collapses context across domains, audiences, and intentions violating key expectations of contextual privacy and making it difficult for individuals to control how they are represented [28].

This paper argues that current privacy frameworks and policies, which largely emphasize data collection, consent, and ownership, are insufficient for addressing the unique risks

posed by LLMs trained on public web data. We contend that the ability of LLMs to derive logical but unverified connections about individuals, which we refer to as “inferential privacy violations,” necessitates a fundamental reevaluation of how privacy is conceptualized in the era of generative AI. We call for a shift toward regulatory and design-based frameworks that center on inference, context, and the lived realities of those most exposed to synthetic profiling.

2 The Rise of Inferential Privacy Harms

Inferential privacy refers to the harms that arise when a system infers sensitive or identifying information about someone based on indirect or non-explicit signals. While the concept has been previously explored in the context of data mining and predictive analytics [7, 21], the generative capabilities of LLMs significantly amplify these concerns by automating and scaling the inference process Figure 2.

Inferential privacy has long been a concern in systems like advertising algorithms, social media analytics, and recommendation engines [18, 37]. These tools often guess personal traits, such as political views or health concerns, based on online behavior or associations with others. But LLMs raise new concerns because of how they present these inferences. Unlike older systems [18] that make predictions silently in the background, internet-enabled LLMs generate fluent, confident natural language outputs that directly state inferences as if they were facts. They pull from a wide mix of sources including social media posts, academic papers, and news articles, and often merge information across different contexts to generate their responses. While tools like ChatGPT may provide citations or links, these references are often incomplete, misleading, or fail to show how different pieces of information were combined. More importantly, LLMs rarely make it clear when they are stating a factual claim versus generating an inference based on patterns in the data. This leads to a

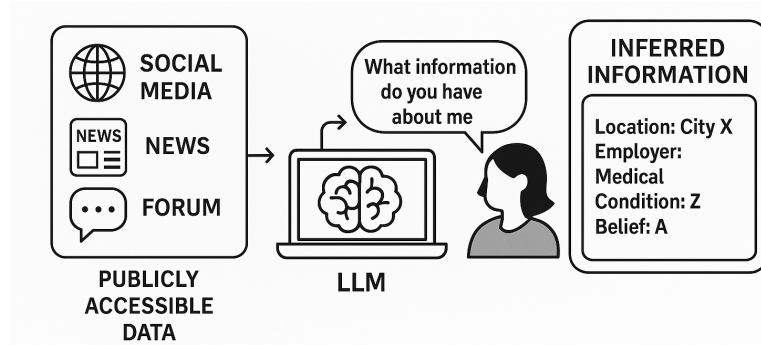


Figure 2: Inferential Privacy Concerns in LLM systems

powerful but risky form of synthesis: one that makes inferences appear factual, coherent, and highly personalized. As a result, users may trust outputs that are speculative or incorrect, increasing the likelihood of harm, especially when the content touches on sensitive aspects of identity or behavior. In the context of LLMs, inferential privacy risks can surface in two troubling ways, each with distinct implications for harm and interpretation:

Accurate but Unwanted Inferences: When models correctly piece together personal details, such as de-anonymizing a pseudonymous user by linking public blog content with a social media handle based on contextual clues distributed across sources. Prior studies have shown how minimal cross-referenced data points can be enough to re-identify individuals in supposedly anonymized datasets [19, 35]. For instance, someone may write anonymously on a mental health forum but also maintain a personal blog under their real name. An LLM might synthesize writing patterns, shared location tags, or similar content to connect the two identities revealing sensitive information the person had tried to keep separate. LLMs trained on broad and unfiltered corpora are especially vulnerable to performing such linkages without user intent or awareness [40].

Inaccurate but Plausible Inferences: When models hallucinate connections, such as falsely attributing a mental health diagnosis or political stance to an individual. These outputs may not only be untrue but also damaging, especially when users treat LLM outputs as authoritative. Research on hallucinations in language models highlights how syntactically fluent but semantically incorrect outputs are common and can mislead users [23, 29]. For example, imagine an international student in the U.S. who has posted publicly on Twitter about visa delays and shared an article critical of U.S. immigration policy on LinkedIn. When someone queries an LLM about that student, the model might infer and generate a response suggesting they are “at risk of deportation” or “facing immigration issues.” Even though the student has a valid visa and no history of legal trouble, the model’s confident language makes the output seem factual. This kind of speculative

inference can lead to reputational harm, discrimination, or anxiety especially in contexts like job applications, classroom environments, or social circles

Both types of inference risks carry significant implications. Individuals may face reputational damage, social stigma, doxxing, or harassment, particularly when LLMs generate and surface information that appears factual but is either speculative or taken out of context. Unlike traditional privacy violations, these harms do not result from unauthorized data access or leaks; they stem from the model’s ability to generate new, inferred knowledge based on partial public inputs [13]. Moreover, these risks are magnified in high-stakes contexts. For example, in mental health, criminal justice, or education, even a speculative inference, such as suggesting that someone has a criminal background or health condition, could result in discrimination or psychological harm [36, 44]. The logic of inference itself thus becomes a critical site of privacy harm: one that falls outside the boundaries of traditional compliance frameworks but carries profound real-world consequences.

3 The Inadequacy of Existing Privacy Frameworks

3.1 Why Current Laws and Tools Fall Short:

Current legal and ethical frameworks, such as the General Data Protection Regulation (GDPR) [2], the California Consumer Privacy Act (CCPA) [1], and emerging global standards, emphasize foundational principles like informed consent, purpose limitation, data minimization, and user access and control. While these measures have been pivotal in creating accountability for how personal data is collected, processed, and shared, they are rooted in assumptions that data is static, explicitly collected, and individually attributable.

However, inferential outputs generated by LLMs do not neatly fit these categories. In many cases, individuals are neither aware that their data is part of training corpora nor that these models may synthesize additional information about them. Because these systems operate probabilistically and generatively [27], the harms emerge not from direct data col-

lection, but from the outputs they produce outputs which may combine disparate signals into new, sensitive narratives [44].

Legal scholars have begun to note the limits of consent-based privacy regimes in the age of predictive and generative analytics [38, 42]. These frameworks provide little recourse when models construct new profiles or characteristics without collecting those attributes explicitly. Moreover, technical claims made by LLM developers that models do not memorize exact data or that their training data was "publicly available" obscure the social and psychological consequences of inferred information [42, 44].

From a normative perspective, focusing only on the legality of scraping publicly accessible content sidesteps deeper ethical obligations. Public availability does not equate to contextually appropriate use, and privacy harms are often shaped by how information is interpreted or repurposed, not merely by its origin [39]. Therefore, current frameworks fall short not just because they lack coverage of inferential harms, but because they rely on outdated notions of informational flow and user agency [33]. If LLMs can construct private identities or fictionalized profiles without input or control from the individuals involved, then privacy regulation must expand to include these generative risks.

3.2 What Differential Privacy Framework Covers and What It Misses:

Differential privacy is a well-established method designed to protect individuals in large datasets by ensuring that their presence or absence cannot be confidently determined from statistical outputs [16]. This framework has shaped how many researchers and developers think about privacy, defining it primarily in terms of whether an individual's data is explicitly included and whether it can be linked to or identified.

However, differential privacy frameworks often assume that if a model generates information about a person who was not included in the training dataset, then no harm has occurred [14]. This paper challenges that assumption. We argue that individuals can still be harmed by what a model infers about them, especially when those inferences reflect patterns learned from large populations and projected onto others.

For example, even if you never posted publicly about your health, a model might suggest that you belong to a particular risk group based on your location, age, and the behavior of people like you in the training data. This kind of projection may feel personal, even if your specific data was never used. In this way, differential privacy addresses one kind of risk (being re-identified), but not another (being misrepresented, stereotyped, or profiled based on how the model generalizes). Our focus is on that second kind of harm, oftentimes neglected by privacy frameworks based on the assumption of differential privacy. We argue that privacy protections need to be expanded to consider these emergent risks of inference, not just inclusion.

4 Disproportionate Impact on Vulnerable Communities

Inferential privacy risks are not evenly distributed. Marginalized and vulnerable populations, including youth, LGBTQ+ individuals, activists, people with disabilities, and those in politically unstable or global south regions, are disproportionately affected by automated inference systems [31, 32, 36, 41]. Many individuals in these groups rely on pseudonymity, anonymity, or curated digital identities as a form of self-protection and boundary management [5, 22].

When LLMs aggregate fragmented online traces, such as combining usernames, metadata, and context clues, they can compromise this protective layer, exposing sensitive associations or personal details. Studies on data surveillance and re-identification risks demonstrate how minor data fragments can be leveraged to triangulate identities, especially for those who are already over-surveilled or socially marginalized [22, 35].

Furthermore, LLMs are trained on data that is often culturally, linguistically, and politically biased. Without adequate oversight, they may reproduce stereotypes or erase context. For instance, research shows that generative models often fail to accurately represent disabled people and queer communities, either omitting their experiences or misrepresenting them through outdated, pathologizing, or stigmatizing language [22, 25].

Consider a young queer activist in a restrictive country: an LLM might incorrectly infer their real identity from subtle indicators in public content, potentially exposing them to legal or social harm. Similarly, a disabled creator using online platforms might find themselves mischaracterized based on the model's exposure to outdated or ableist terminology. These are not merely theoretical risks; they reflect the ways LLMs can amplify systemic inequalities while appearing neutral or authoritative [12].

Because these communities often have limited resources to contest misrepresentations or engage in legal redress, the privacy harms they experience are compounded by power asymmetries. Addressing inferential privacy requires centering these lived experiences, not only in model evaluation but also in governance and design.

5 Reframing Privacy for the LLM Era: A Call to Action

We argue for a paradigm shift: from treating privacy as the protection of static data to recognizing it as the right to not be misrepresented, profiled, or exposed through automated inference [39]. This reconceptualization demands both regulatory innovation and technical reimagining.

Inferential Privacy Audits: There is a pressing need for systematic tools to assess whether LLMs can infer or fabricate sensitive attributes about individuals [40]. We define

inferential privacy audits as evaluations of a model’s tendency to generate identity-related, behavioral, or stigmatizing content about individuals based on minimal inputs. Audits can include red-teaming exercises using structured prompts designed to elicit sensitive attributes (e.g., sexuality, religion, health), benchmarked against metrics such as inference frequency, plausibility, and harm potential [40]. Validating such audits requires collaboration with domain experts and affected users to assess when plausible inferences cross ethical or psychological boundaries [34]. These evaluations should center on populations most vulnerable to being misrepresented and include community-defined criteria for harm and visibility.

Participatory Red-Teaming: Recent research advocates for participatory methods to evaluate algorithmic systems, emphasizing the inclusion of communities most at risk [15, 32]. Red-teaming exercises should extend beyond consultations on security or adversarial testing to meaningfully engage stakeholders and ground GenAI systems in lived experiences.

Context-Aware Design: LLMs must provide contextual transparency to end users, indicating when outputs are probabilistic or inferred rather than factual. Building on HCI principles of explainability and interpretability [17], systems should include affordances that allow users to question, flag, or correct speculative statements. While Nissenbaum’s theory of contextual integrity [30] highlights the need for role- and context-sensitive information flows, the architecture of LLMs inherently collapses these distinctions. We propose design interventions that mitigate context collapse, such as model-output annotations indicating inference vs. fact, tunable output constraints based on domain or risk sensitivity, and participatory flagging systems that allow users to contest or report context-violating generations.

Policy Innovation: Existing laws must be updated to include inference as a distinct category of data-related harm. Legal scholars have proposed expanding the scope of privacy rights to include protections against algorithmic profiling and reputational damage from synthetic content [42]. Regulating inference as a site of harm, regardless of whether the underlying data was legally collected, would help shift the focus from ownership to accountability.

Together, these directions form a blueprint for a rights-based and community-informed approach to AI governance. Rather than retrofitting outdated policies, the moment calls for an expansion of privacy frameworks that account for the power of inference and the relational nature of harm.

6 Conclusion

As generative AI becomes increasingly embedded in public and private life, we must confront an urgent reality: the most profound privacy harms may stem not from what models memorize, but from what they infer. These inferences, which are plausible, synthetic, and often unverifiable, can misrepresent, de-anonymize, or profile individuals in ways that evade

current legal frameworks and technical safeguards. Inference is not neutral, nor is it harmless. When systems fabricate connections and identities, the line between fiction and fact, public and private, collapses. This paper has argued that existing privacy paradigms, grounded in consent, control, and data ownership, are inadequate for addressing the generative, probabilistic, and often opaque operations of LLMs. We introduced the concept of inferential privacy harms, analyzed their disproportionate impact on marginalized communities, and proposed a roadmap of regulatory and design innovations to mitigate these risks. To move forward, privacy must be reframed not just as a right to restrict data access, but as a right to dignity, contextual integrity, and protection from algorithmic misrepresentation. Only then can we ensure that LLMs become tools of empowerment rather than instruments of exposure and epistemic harm.

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